Quantitative Shift-One Shadows of the Determinant-Two Frame

Exact Even-Carrier Transfer and the Active Odd-Carrier Firewall

Liang Wang

July 2026

Abstract

The unrestricted shift-two Liouville sign has an exact quantitative shadow on its even carrier:

$$\lambda(2m-2)\lambda(2m) = \lambda(m-1)\lambda(m).$$

Consequently, proved quantitative logarithmic and almost-all-scale results for the consecutive shift-one correlation transfer to that even subcarrier. This does not estimate the active determinant-two TPC packet. On the simultaneous squarefree support of TPC-127, the affine values D, V are odd, and hence $n = sV = aD + 2$ is odd. We prove a complementary determinant invariant: on any residue progression contained in the odd integers, extracting all fixed contents from $n-2, n$ leaves an affine determinant divisible by four. It cannot produce the determinant-one pair $t, t+1$. We cite Helfgott–Radziwiłł and Pilatte only for their stated shift-one conclusions. Separately, Tao–Teräväinen now provide a quantitative almost-all-scale theorem uniformly for positive affine data of sufficiently small polylogarithmic height. That new corridor can cover eligible determinant-two affine components, but it does not supply the squarefree component reassembly, deterministic all-prefix selector, or physical return. The transfer and firewall are exact L0 statements; the full actual packet remains open, and no positive L2 result is claimed.

1 Four theorem scopes that must not be merged

Let $\lambda(n) = (-1)^{\Omega(n)}$. There are three distinct claims in play.

Fixed affine, qualitative logarithmic. Tao proved logarithmically averaged two-point cancellation for arbitrary fixed nonparallel affine forms [3]. TPC-137 uses this theorem to close the full squarefree masks for fixed determinant-two data [6].

Consecutive shift one, quantitative. Helfgott and Radziwiłł proved, among other statements, a quantitative logarithmic bound for $\lambda(n)\lambda(n+1)$ and ordinary cancellation at almost all scales [1]. Pilatte later proved

$$\sum_{n \le x} \frac{\lambda(n)\lambda(n+1)}{n} \ll (\log x)^{1-c} \tag{1}$$

for some absolute $c > 0$ [2].

Small-polylog affine, quantitative almost all scales. Tao and Teräväinen prove the following uniform special case of their quantitative correlation theorem: for some sufficiently small absolute $c_0 > 0$, outside a set of scales of logarithmic density $O((\log X)^{-c_0})$,

$$\frac{1}{N} \sum_{n \le N} \lambda(a_1 n + b_1)\lambda(a_2 n + b_2) \ll (\log N)^{-c_0} \tag{2}$$

uniformly for positive integers

$$a_1, a_2, b_1, b_2 \le (\log N)^{c_0}, \qquad a_1 b_2 \ne a_2 b_1$$

[4, Theorem 3.1 and Remarks 3.2]. Shrinking c_0 absorbs harmless differences between the height and saving exponents. This is a genuine arbitrary-affine quantitative corridor, but only at nonexceptional scales and only inside its stated small-polylog range.

Actual determinant two. TPC-127 produces

$$n = sV(z) = aD(z) + 2$$

on one generally growing progression, behind quotient-squarefree masks, physical weights, phases and prefixes [5]. Neither (1) nor the stated shift-one almost-scale result is, as cited here, a theorem on this actual family. Equation (2) can estimate an individual reduced component only after its literal coefficients, constants and CRT modulus have been proved to lie in the small-polylog corridor; it still leaves exceptional-scale selection and complete reassembly.

Definition 1.1 (Source-safe quantitative gate)**.** A quantitative determinant-two affine theorem is a separately proved estimate whose statement itself includes the desired determinant-two affine forms and all required growth quantifiers. A shift-one theorem plus an unproved assertion that its method "should extend" does not pass this gate. The proved Tao–Teräväinen corridor passes only for records satisfying its explicit height and nonexceptional-scale hypotheses.

2 The exact even-carrier transfer

Proposition 2.1 (Shift two becomes shift one on even integers)**.** *For every $m \ge 2$,*

$$\boxed{\lambda(2m-2)\lambda(2m) = \lambda(m-1)\lambda(m).} \tag{3}$$

Consequently, for any finite interval J,

$$\sum_{\substack{n \in 2J \\ n \text{ even}}} \frac{\lambda(n-2)\lambda(n)}{n} = \frac{1}{2} \sum_{m \in J} \frac{\lambda(m-1)\lambda(m)}{m}. \tag{4}$$

The corresponding unweighted interval sums agree after the same change of variable, up to endpoint conventions.

Proof. Complete multiplicativity gives

$$\lambda(2m-2)\lambda(2m) = \lambda(2)^2 \lambda(m-1)\lambda(m),$$

and $\lambda(2)^2 = 1$. Substitute $n = 2m$ to obtain the two sum identities. □

Corollary 2.2 (What the quantitative source theorems control)**.** *Every stated shift-one logarithmic or almost-all-scale estimate transfers, with its native quantifiers and rate, to the even subcarrier of the unrestricted sequence $\lambda(n-2)\lambda(n)$.*

Proof. Apply proposition 2.1; the fixed dilation by two changes only absolute constants and interval endpoints. □

This is genuine arithmetic information, but its carrier must be checked before it is inserted in H3.

3 The actual squarefree carrier is odd

TPC-127 proves the following arithmetic geometry. If

$$D(z) = d + sz, \qquad V(z) = u + az, \qquad su - ad = 2,$$

with a, s odd, then $D(z), V(z)$ have the same parity. On their simultaneous squarefree support, the even–even case is impossible.

Proposition 3.1 (Empty overlap). *On the simultaneous squarefree determinant-two support,*

$$D(z) \equiv V(z) \equiv n(z) \equiv 1 \pmod 2, \qquad n(z) = sV(z) = aD(z) + 2.$$

Hence the carrier controlled by corollary 2.2 has empty intersection with the active squarefree carrier.

Proof. The oddness and coprimality of D, V are the parity classification of TPC-127. Since s is odd, $n = sV$ is odd. The carrier in corollary 2.2 consists of even n. □

Remark 3.2 (The positive result is not discarded). Proposition 3.1 does not weaken the shift-one theorems. It says that their exact even-carrier consequence bounds a different subsequence. Calling it an estimate of the odd squarefree carrier would be a support error.

4 An odd-carrier determinant invariant

The possibility remains that a residue change and extraction of fixed contents might convert the odd carrier to consecutive forms. The next theorem rules out that elementary move.

Theorem 4.1 (Even determinant survives odd-carrier reduction). *Let M be even and r odd, so that $n = r + Mt$ is odd for every integer t. Put*

$$c_- = (M, r-2), \qquad c_+ = (M, r),$$

and define the primitive affine forms

$$L_-(t) = \frac{Mt + r - 2}{c_-}, \qquad L_+(t) = \frac{Mt + r}{c_+}.$$

Then c_-, c_+ are odd and coprime, and

$$\boxed{\det(L_-, L_+) = \frac{2M}{c_-c_+} \in 4\mathbb{Z}.} \tag{5}$$

In particular, this pair cannot equal, up to order and fixed Liouville factors, the primitive consecutive pair $t + b, t + b + 1$, whose determinant has absolute value 1.

Proof. Both r and $r - 2$ are odd, so both contents are odd. Furthermore

$$(c_-, c_+) \mid (r - 2, r) \mid 2,$$

and hence the contents are coprime. Since each divides M, their product divides M. Each displayed affine form is primitive because its content has been divided out. The determinant of the coefficient pairs is

$$\frac{M}{c_-}\frac{r}{c_+} - \frac{M}{c_+}\frac{r-2}{c_-} = \frac{2M}{c_-c_+}.$$

Because M is even and c_-c_+ is odd, $M/(c_-c_+)$ is even; hence the determinant is divisible by four. The determinant of $(1, b), (1, b + 1)$ is 1. □

Remark 4.2 (Scope of the obstruction). Theorem 4.1 forbids a residue/content relabeling, not a new analytic proof. One may try to rebuild an expansion or entropy argument directly for determinant two. That effort is precisely the open gate in definition 1.1.

5 Finite diagnostic and claim boundary

The deterministic script verifies (3) term by term, enumerates simultaneous squarefree determinant-two samples, and checks primitivity, content coprimality and (5) over many odd residue classes. These are exact regressions. It may also report finite correlations, but no fitted decay is used as an asymptotic premise.

Statement	Level and status
Even shift-two to shift-one transfer	Exact theorem (L0).
Quantitative consequence on the even carrier	Proved external input with unchanged shift-one scope.
Active determinant-two carrier is odd and disjoint	Exact support theorem (L0).
Odd-carrier reduced determinant remains even	Exact firewall (L0).
Quantitative logarithmic or almost-scale theorem on active determinant two	Proved for eligible small-polylog affine data outside a small log-density exceptional scale set; actual-family return open.
Growing physical all-prefix power estimate	Open L2 target.

Thus TPC-138 identifies a real but ineligible quantitative shadow. The next task is not another parity relabeling: it is to compare the literal squarefree/periodic decomposition with the proved small-polylog affine corridor, account for all component losses, and prevent deterministic prefixes from concentrating on its exceptional scales.

References

[1] Harald Andrés Helfgott and Maksym Radziwiłł. Expansion, Divisibility and Parity, 2021.

[2] Cédric Pilatte. Improved Bounds for the Two-Point Logarithmic Chowla Conjecture, 2023.

[3] Terence Tao. The Logarithmically Averaged Chowla and Elliott Conjectures for Two-Point Correlations. *Forum of Mathematics, Pi*, 4:e8, 2016. doi: 10.1017/fmp.2016.6.

[4] Terence Tao and Joni Teräväinen. Quantitative Correlations and Some Problems on Prime Factors of Consecutive Integers, 2026.

[5] Liang Wang. Determinant-Two Liouville Pullback. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-127-determinant-two-liouville-pullback, 2026.

[6] Liang Wang. Prime-Square Closure of the Frozen Determinant-Two Logarithmic Shadow, 2026. TPC-137 manuscript.